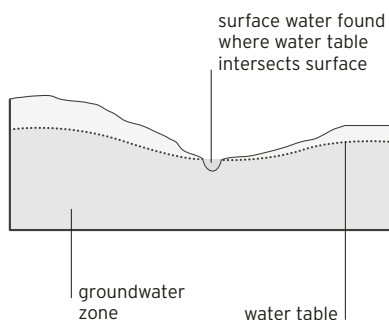


GROUNDMASS The fine-grained material in which larger crystals or fragments are embedded in igneous (and some sedimentary) rocks. See also *matrix*.

GROUNDWATER ▼ Water held in spaces (interstices) between the grains of rocks underground. The water table represents the top of the groundwater zone. See also *aquifer*, *water table*.



GUYOT A flat-topped, undersea volcanic mountain, or seamount. The summit of a guyot is more than 660 ft (200 m) below sea level. See also *seamount*.

GYRE The large-scale rotation of an ocean current. See also *current*.

H

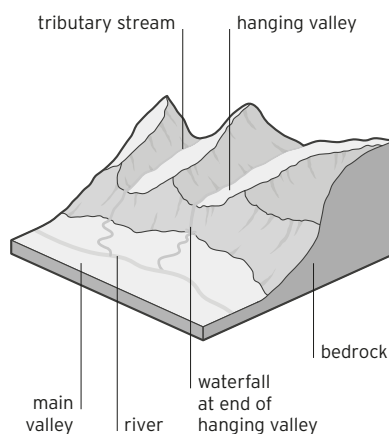
HABITAT Any area that can support a particular group of plants or animals.

HALO A hazy ring around the Moon or Sun, which is caused by the refraction of light as it passes through high clouds.

HAMADA A flat, rocky type of desert terrain, mainly devoid of sand. See also *desert pavement*.

HAMMOCK A stand of trees growing on raised ground above a wetland. Hammocks are common in the Florida Everglades.

HANGING VALLEY ▼ A tributary valley that joins a main valley high above the floor of the latter. Hanging valleys usually occur where a glacier has deepened the larger valley.



HAWAIIAN ERUPTION The least explosive type of volcanic eruption, with a relatively gentle flow of basaltic lava from a vent.

HEADLAND A narrow area of land projecting from a coastline. Headlands often exist because they are made of harder rocks than the coast on either side.

HEADWATER The upper portion of any stream or river, close to its source.

HOT SPRING The emergence of hot water and steam from the ground. Hot springs occur when hot rocks beneath the surface have heated groundwater. See also *geyser*.

HOTSPOT An area of volcanic activity thought to originate deep in Earth's mantle. Tectonic plates passing over hotspots are marked by lines of volcanoes that become older with increased distance from the hotspot. See also *mantle plume*.

HUMUS A dark-colored component of soil made up mainly of decomposed plant material.

HURRICANE A large weather system that develops and moves over tropical and subtropical oceans. Hurricanes feature destructive winds circulating rapidly around an area of low pressure, torrential rain, and storm surges. Also called a tropical cyclone or a typhoon. See also *cyclone*, *latent heat*.

HYDROTHERMAL VEIN Any concentration of minerals formed by the precipitation of solids from hot, mineral-laden water. See also *mineral*, *vein*.

HYDROTHERMAL VENT An outlet for water that has been heated by molten or hot rocks. Hydrothermal vents that emit hot, dark water from the ocean floor are called black smokers. See also *black smoker*, *geyser*, *hot spring*, *white smoker*.

I

ICE AGE A period of long-term reduction in global temperatures that results in the expansion of polar and continental ice-sheets and glaciers. See also *glacier*, *ice sheet*.

ICE CAP An ice mass covering a large area, but less than 19,000 square miles (50,000 square km). An ice cap's form overrides underlying landscape features. See also *glacier*, *icefield*, *ice sheet*.

ICE LEAD A large fracture in an expanse of sea-ice. See also *sea-ice*.

ICE SHEET A layer of ice covering an area greater than 19,000 square miles (50,000 square km) for

a prolonged period. See also *ice cap*, *icefield*.

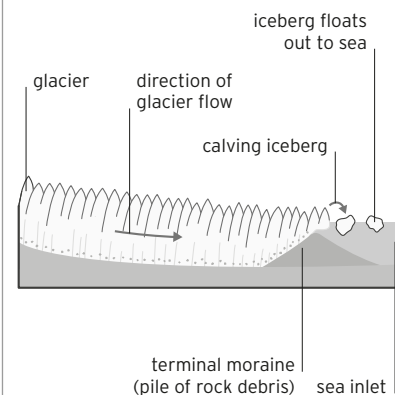
ICE SHELF A large area of floating ice that is permanently attached to land. The Ross Ice Shelf, attached to Antarctica, is the world's largest.

ICE STREAM Part of an ice sheet that is moving faster than the surrounding ice. Ice streams are common in Antarctica. See also *ice sheet*.

ICE WEDGE A vertical, wedge-shaped mass of ice in the soil in a periglacial area. See also *ice-heave*, *periglaciation*.

ICE-HEAVE Disruption of the land surface due to the growth of ice within the soil. Ice-heave is one of the features of periglacial regions. See also *ice wedge*, *periglaciation*, *pingo*.

ICEBERG ▼ A floating mass of ice that has calved from a glacier, ice sheet, or ice shelf. See also *calving*, *glacier*, *ice shelf*, *ice sheet*.



ICEFIELD A large expanse of ice whose surface shape and extent is determined by underlying landscape features. Icefields are often interconnected valley glaciers from which higher mountain peaks rise. See also *glacier*, *ice cap*, *nunatak*.

IGNEOUS INTRUSION See panel, right.